

REFLOWECONOMY · DESIGN DECISION RECORD

ReflowEconomy TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	RFE-DDR-001	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/refloweconomy		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 5, and item 6 for the interim); item 7, the license for the first release (item 6) and the new TRL 3 items 8 to 11 remain proposed, awaiting Amish

Safety: The reference micro-factory has a shredder, surfaces at 190 to 200 °C, melt fumes, water near electricity, lithium cells in the waste stream and a high fire load. None of the decisions below relaxes a safety rule. Item 1 keeps molten aluminium out of the reference floor until an add-on bay has its own safety case.

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed seven items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." The same instruction approved cross-cutting SwapCell interface items, a pricing rule for shared SwapCell packs, and a rule that community designs pick co-design partners per area later.

This record lists what that instruction decides and what it leaves open because there was no single recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 section) and RFE-PRC-001 v0.2. They are not repeated here, except for the open items.

■ Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Reference line	Decided by Amish, 2026-09-25: go with recommendation. Option (a): plastics only. Aluminium is a separate add-on bay with its own safety case, not part of the reference floor	RFE-PRC-001 v0.3, RFE-REQ-001 v0.3 R2 (redefined), docs/playbook/recipes/aluminium.md
2	Footprint	Decided by Amish, 2026-09-25: go with recommendation. Option (a): two 40 ft container footprints or a shed of about 60 m ²	cad/src/model.py , RFE-DWG-001, RFE-CAL-001 section 8
3	PET route	Decided by Amish, 2026-09-25: go with recommendation. Option (a): PET is sold as clean flake, not processed in house	docs/playbook/recipes/pet-flake.md , RFE-REQ-001 assumptions
4	Passport schema v0.2	Decided by Amish, 2026-09-25: go with recommendation. Add parent_batch_ids , schema_version , product_form and a recycled content statement aligned with ISO 14021, and publish example records, at TRL 3. Issued as a new file; v0.1 is kept unchanged	standards/material-passport-v0.2.schema.json , standards/examples/ , RFE-REQ-001 R13 and R14
5	Electrical supply	Decided by Amish, 2026-09-25: go with recommendation. Option (b): design for single-phase with staggered heating, so the playbook works on weak grids. Implemented as a heater interlock (press heaters locked out while the shredder or extruder runs) on a 230 V, 40 A supply	RFE-CAL-001 section 5, bom/bom.csv item 11, RFE-REQ-001 assumptions
6	Licensing of the text	Decided by Amish, 2026-09-25: go with recommendation. Keep the playbook text under CERN-OHL-S-2.0 with the repo for now. No license was changed. The choice of license for the first release remains open (see below)	This record

The pitch and problem line had no recommended rewording in the TRL 2 review and are unchanged. budget_usd stays null: this is a playbook repo, and the TRL 2 review made no budget recommendation. The reference equipment list is checked against R10 instead.

Cross-cutting approvals from the same instruction, recorded here:

- **SwapCell interface v0.3.** Decided by Amish, 2026-09-25 (cross-cutting): a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles. ReflowEconomy does not use SwapCell, so no v0.3 items apply.
- **Shared packs are priced once.** Decided by Amish, 2026-09-25 (cross-cutting). Not applicable: ReflowEconomy has no SwapCell pack.
- **Co-design partners per area later.** Decided by Amish, 2026-09-25 (cross-cutting): community designs pick co-design partners per area later. Item 7 stays open under this rule.

Items that remain open

These stay **Proposed**, awaiting Amish.

#	Item	Options	Recommendation
6 (release)	License for the first release of the playbook text	(a) keep CERN-OHL-S-2.0 for everything; (b) CC BY-SA 4.0 for documents, CERN-OHL-S-2.0 for hardware and MIT for scripts	(b), decided before the first release; this is a licensing decision for Amish
7	First co-design partner	(a) a waste-picker cooperative; (b) an existing Precious Plastic workspace; (c) a municipal program	(a) or (b); left open under the cross-cutting rule that partners are picked per area later
8	R8 definition (new at TRL 3)	(a) keep "residue to licensed disposal 20 % or less" and count process losses (not met at 28.4 %); (b) apply the 20 % limit to sorting residue only and report process losses separately; (c) keep the target and set a minimum input quality at intake	(c) with (a): keep the honest total and add an intake quality rule, because the site can refuse poor loads but cannot make losses disappear
9	Single-phase sheet press (new at TRL 3)	(a) a derated 5 kW press with lighter platens (the TRL 3 design, not a published open design); (b) the published 15 kW press on a site with a 400 V supply; (c) buy sheets pressed elsewhere and make only beams	(a) for the playbook, with (b) as an option where three-phase exists; RFE-CAL-001 shows (a) has almost no time margin
10	Fume fan size (new at TRL 3)	(a) 1.1 kW, 91 % loaded with dirty filters; (b) 1.5 kW, which raises the maximum demand to 38 A	(b), because filter loading is hard to police on a small site
11	Structure of the container option (new at TRL 3)	(a) remove the adjoining side walls with engineered reinforcement; (b) keep the walls and cut two wide openings; (c) prefer a rented shed	(c) where available, (a) only with a structural engineer's design

■ Consequences

- RFE-PRB-001, RFE-PRC-001 and RFE-REQ-001 move to v0.3 with these decisions. The design choices in items 1 to 5 are no longer "proposed".
- RFE-REQ-001 R2 is redefined: the aluminium recipe serves the add-on bay, not the reference floor. R13 and R14 now name schema v0.2. The assumptions state the single-phase supply.
- The TRL 3 evidence (RFE-CAL-001, `cad/src/model.py`, RFE-DWG-001 and `bom/bom.csv`) follows these decisions.
- TRL 4 is on hold by Amish's instruction. Nothing in this record starts TRL 4 work.